Computer Network

09 April 2026 22:55

1. WHAT IS A COMPUTER NETWORK?

A computer network is a collection of interconnected devices (computers, printers, servers) that can share data and resources.
Example: Your home Wi-Fi connects your phone, laptop, and TV — that's a network.

2. TYPES OF NETWORKS

Type	Full Form	Range	Example
PAN	Personal Area Network	~10m	Bluetooth, USB
LAN	Local Area Network	Building/Campus	Office, College lab
MAN	Metropolitan Area Network	City	Cable TV network
WAN	Wide Area Network	Country/World	Internet

3. NETWORK TOPOLOGIES

The physical/logical arrangement of devices.

- **Bus** — All devices share one cable. Simple but if cable breaks, all fail.
- **Star** — All devices connect to a central hub/switch. Most common. If hub fails, all fail.
- **Ring** — Devices connected in a circle. Data travels in one direction.
- **Mesh** — Every device connects to every other. Most reliable, most expensive.
- **Tree** — Combination of Star + Bus. Used in large organizations.
- **Hybrid** — Mix of two or more topologies.

Interview Q: *Which topology is most fault-tolerant?* → **Mesh**

4. OSI MODEL (Most Important)

Open Systems Interconnection — 7 layers model for how data travels.
Mnemonic: All **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing (bottom to top)

Layer	Name	Function	Protocol/Device
7	Application	User interface	HTTP, FTP, SMTP, DNS
6	Presentation	Encryption, compression, translation	SSL/TLS, JPEG
5	Session	Start/stop/manage sessions	NetBIOS, PPTP
4	Transport	End-to-end delivery, error control	TCP, UDP
3	Network	Routing, IP addressing	IP, Router
2	Data Link	MAC address, framing, error detection	Ethernet, Switch
1	Physical	Bits over physical medium	Cable, Hub, NIC

Key Points:

- Sender: Data goes **top to bottom** (Layer 7 → 1)
- Receiver: Data goes **bottom to top** (Layer 1 → 7)
- Each layer adds a **header** (encapsulation) when sending, removes it when receiving (decapsulation)

Interview Qs:

- *At which layer does routing happen?* → Layer 3 (Network)
- *Which layer does a switch work on?* → Layer 2
- *Which layer does a router work on?* → Layer 3
- *Where does SSL/TLS work?* → Layer 6 (Presentation)

5. TCP/IP MODEL

Practical model used in real internet. Has 4 layers.

TCP/IP Layer **OSI Equivalent**

5. TCP/IP MODEL

Practical model used in real internet. Has 4 layers.

TCP/IP Layer	OSI Equivalent
Application	Application + Presentation + Session
Transport	Transport
Internet	Network
Network Access	Data Link + Physical

Interview Q: *Difference between OSI and TCP/IP?* → OSI has 7 layers (theoretical), TCP/IP has 4 layers (practical, used in real networks)

6. IP ADDRESS

A unique address given to every device on a network.
IPv4: 32-bit, written as 4 octets → 192.168.1.1
IPv6: 128-bit, written in hex → 2001:0db8:85a3::8a2e:0370:7334

IP Address Classes (IPv4)

Class	Range	Default Subnet Mask	Use
A	1–126	255.0.0.0	Large organizations
B	128–191	255.255.0.0	Medium organizations
C	192–223	255.255.255.0	Small organizations
D	224–239	—	Multicasting
E	240–255	—	Research/Reserved

127.x.x.x → Loopback address (localhost) — used to test your own machine.

Private IP ranges (not routable on internet):

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Interview Q: *What is 127.0.0.1?* → Loopback/localhost — tests if TCP/IP stack is working on your machine.

7. SUBNETTING

Dividing a large network into smaller sub-networks.
Subnet Mask tells which part is network and which is host.
Example:

- IP: 192.168.1.100
- Subnet Mask: 255.255.255.0
- Network part: 192.168.1
- Host part: .100

CIDR Notation: 192.168.1.0/24 → /24 means 24 bits for network → 8 bits for host → 2^8 - 2 = **254 usable hosts** (subtract network address and broadcast address)
Interview Q: *Why do we subtract 2 in subnetting?* → One for network address, one for broadcast address.

8. TCP vs UDP

Feature	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmit)	Unreliable
Speed	Slower	Faster
Order	Data arrives in order	No guarantee
Use Case	HTTP, Email, FTP	DNS, Video streaming, Gaming

TCP 3-Way Handshake (Very Important!)

1. **SYN** — Client sends "I want to connect"
2. **SYN-ACK** — Server replies "OK, ready"
3. **ACK** — Client confirms "Let's go"

Interview Q: *Why is UDP used for video streaming despite being unreliable?* → Speed matters more than perfect delivery. Missing a few frames is OK, but delay is not.

9. IMPORTANT PROTOCOLS + PORT NUMBERS

9. IMPORTANT PROTOCOLS + PORT NUMBERS

Protocol	Port	Use
HTTP	80	Web browsing
HTTPS	443	Secure web browsing
FTP	20, 21	File transfer
SSH	22	Secure remote login
Telnet	23	Unsecure remote login
SMTP	25	Sending email
DNS	53	Domain name resolution
DHCP	67, 68	IP address assignment
POP3	110	Receiving email
IMAP	143	Receiving email (sync)
SNMP	161	Network management
RDP	3389	Remote Desktop

Interview Q: *Difference between HTTP and HTTPS?* → HTTPS uses SSL/TLS encryption, HTTP does not. HTTPS = HTTP + Security.

10. DNS (Domain Name System)

Converts domain names to IP addresses.
Example: You type google.com → DNS converts it to 142.250.x.x → browser connects.

DNS Resolution Steps:

1. Check **browser cache**
2. Check **OS cache** (hosts file)
3. Ask **Recursive Resolver** (your ISP's DNS)
4. Ask **Root Name Server** → tells which TLD server
5. Ask **TLD Server** (.com, .in) → tells authoritative server
6. Ask **Authoritative Server** → gives final IP

Interview Q: *What happens when you type www.google.com in a browser?* → DNS lookup → TCP connection → HTTP/HTTPS request → Server responds → Browser renders

11. DHCP (Dynamic Host Configuration Protocol)

Automatically assigns IP address, subnet mask, gateway, DNS to devices.
Process (DORA):

- **D** — Discover (client broadcasts "anyone there?")
- **O** — Offer (server offers an IP)
- **R** — Request (client says "I want that IP")
- **A** — Acknowledge (server confirms)

Interview Q: *What is the difference between static and dynamic IP?* → Static = manually assigned, never changes. Dynamic = assigned by DHCP, can change.

12. ARP (Address Resolution Protocol)

Converts IP address → MAC address on a local network.
Example: You know IP 192.168.1.5 but need its MAC to send data. ARP broadcasts "Who has 192.168.1.5?" → that device replies with its MAC.
RARP — Reverse ARP → MAC to IP (old, replaced by DHCP)

13. MAC ADDRESS

- 48-bit hardware address burned into NIC
- Written as: 00:1A:2B:3C:4D:5E
- First 3 bytes = **OUI** (manufacturer ID)
- Last 3 bytes = **device ID**
- MAC works at **Layer 2**, IP works at **Layer 3**

14. NETWORKING DEVICES

Device	Layer	Function
Hub	L1	Broadcasts data to all ports (dumb)
Switch	L2	Sends data only to correct MAC (smart)
Router	L3	Routes data between networks using IP
Bridge	L2	Connects two LAN segments
Gateway	L4-L7	Connects networks with different protocols
Modem	L1	Modulates/demodulates signal (analog ↔ digital)
NIC	L1/L2	Network Interface Card — connects device to network
Firewall	L3-L7	Filters traffic based on rules

Interview Q: *Difference between Hub and Switch?* → Hub broadcasts to ALL ports. Switch sends only to the destination MAC port. Switch is more efficient and secure.

15. FIREWALL

Monitors and controls incoming/outgoing traffic based on security rules.
Types:

- **Packet Filter** — Checks IP/port, fast but basic
- **Stateful** — Tracks connection state
- **Application Layer** — Inspects content (deep inspection)
- **Proxy Firewall** — Acts as middleman

16. NAT (Network Address Translation)

Converts **private IP → public IP** when accessing the internet.
Example: 100 devices at home all have private IPs (192.168.x.x). Your router has 1 public IP. NAT maps all internal requests to that one public IP.
Types:

- **Static NAT** — 1 private IP → 1 public IP
- **Dynamic NAT** — pool of public IPs
- **PAT (Port Address Translation) / NAT Overload** — many private IPs → 1 public IP using ports (most common)

17. VPN (Virtual Private Network)

Creates an **encrypted tunnel** over the public internet — makes you appear to be on a private network.
Used for: Remote work, Privacy, Bypassing geo-restrictions.

18. HTTP vs HTTPS

- **HTTP** — HyperText Transfer Protocol, Port 80, no encryption
- **HTTPS** — HTTP + SSL/TLS, Port 443, encrypted
- **SSL handshake** happens before data transfer in HTTPS

19. NETWORK SECURITY CONCEPTS

Common Attacks:

- **DoS/DDoS** — Flood server with requests to make it crash
- **Man-in-the-Middle (MITM)** — Attacker intercepts communication
- **Phishing** — Fake website/email to steal credentials
- **SQL Injection** — Injecting SQL into input fields
- **ARP Spoofing** — Fake ARP replies to intercept traffic
- **DNS Spoofing/Poisoning** — Fake DNS replies to redirect users
- **Replay Attack** — Capturing and re-sending valid data packets

Defense:

- Encryption (SSL/TLS)
- Firewalls
- IDS/IPS (Intrusion Detection/Prevention)
- VPN
- Strong Authentication (MFA)

20. IMPORTANT CONCEPTS — QUICK FIRE

Bandwidth — Maximum data transfer rate (bits per second)
Latency — Delay in data transmission (ms)
Throughput — Actual data transferred per second Jitter — Variation in latency (bad for video calls) Packet — Unit of data at Network Layer Frame — Unit of data at Data Link Layer Segment — Unit of data at Transport Layer Bit — Unit of data at Physical Layer
Half Duplex — Data flows in one direction at a time (walkie-talkie)
Full Duplex — Data flows both directions simultaneously (phone call)
Simplex — Data flows only one way (TV broadcast)

21. ROUTING PROTOCOLS

Protocol	Type	Algorithm
RIP	Distance Vector	Bellman-Ford, max 15 hops
OSPF	Link State	Dijkstra's algorithm
BGP	Path Vector	Used on the Internet between ISPs
EIGRP	Hybrid	Cisco proprietary

Interview Q: *What is the difference between RIP and OSPF?* → RIP uses hop count (max 15), slow convergence. OSPF uses cost/bandwidth, faster, scalable.

22. SOCKET & PORT

- **Socket** = IP address + Port number → 192.168.1.1:80
- **Port** = logical endpoint for a process
- Ports 0–1023 → Well-known (reserved)
- Ports 1024–49151 → Registered
- Ports 49152–65535 → Dynamic/Ephemeral

23. TOP INTERVIEW QUESTIONS

Q1. What is the difference between TCP and UDP? → TCP is reliable, connection-oriented, ordered. UDP is fast, connectionless, unreliable. TCP for HTTP/email, UDP for video/DNS.
Q2. Explain the OSI model. → 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application. Each layer has a specific role in network communication.
Q3. What happens when you type google.com? → Browser checks cache → DNS resolution → TCP handshake → HTTP request → Server responds → Page renders.
Q4. What is a subnet mask? → Identifies which part of IP is network and which is host. E.g., /24 = 255.255.255.0.
Q5. Difference between Switch and Router? → Switch works at L2, connects devices in

Presentation, Application. Each layer has a specific role in network communication.
Q3. What happens when you type google.com? → Browser checks cache → DNS resolution → TCP handshake → HTTP request → Server responds → Page renders.
Q4. What is a subnet mask? → Identifies which part of IP is network and which is host. E.g., /24 = 255.255.255.0.
Q5. Difference between Switch and Router? → Switch works at L2, connects devices in same network using MAC. Router works at L3, connects different networks using IP.
Q6. What is DHCP and how does it work? → Auto-assigns IPs. Works via DORA: Discover, Offer, Request, Acknowledge.
Q7. What is NAT? → Translates private IPs to public IP. Allows multiple devices to share one public IP.
Q8. What is a firewall? → Security device that monitors and filters network traffic based on rules.
Q9. What is the difference between IPv4 and IPv6? → IPv4: 32-bit, ~4.3 billion addresses. IPv6: 128-bit, virtually unlimited addresses.
Q10. What is a MAC address? → 48-bit hardware address of NIC. Works at Layer 2. Used for communication within a network.
Q11. What is ARP? → Resolves IP to MAC address. Broadcasts on local network to find MAC for a given IP.
Q12. What port does HTTPS use? → 443.
Q13. What is DNS? → Domain Name System. Converts human-readable domain names to IP addresses.
Q14. Difference between Hub, Switch, Router? → Hub broadcasts (L1), Switch uses MAC (L2), Router uses IP (L3).
Q15. What is the loopback address? → 127.0.0.1 — used to test your own machine's network stack.